

Power BI ↔ BigQuery: connecting your Gold marts

The last-mile guide from Day 2 (Tue Oct 6, ~3:05 PM CT), written to keep: how to point your existing Power BI reports at BigQuery Gold marts — desktop connection, the Import vs DirectQuery decision, service-side credentials, and the gateway answer for the M&O team.

Why this matters for your migration

Your migration's endgame is one sentence: **your reports don't change, their source does.** The state's Power BI estate stays exactly as it is — same report authors, same workspaces, same consumers. The only thing that changes is where "Get Data" points. This guide is that change, end to end.

Before you start

- **Power BI Desktop, June 2021 or newer.** Google sign-in from Power BI requires it (Google discontinued embedded-browser sign-ins in 2021).
- **Your own Google account**, with access to the GCP project that holds the Gold dataset. To connect and query, your account needs roles/bigquery.dataViewer on the Gold dataset and roles/bigquery.jobUser on the project — your GCP admin grants both. If you can query the mart in the BigQuery console, you can connect to it from Power BI.
- **Know your target's full name:** project → dataset → table — the same hierarchy you used in the console all course long.
- Reference doc to bookmark: Microsoft keeps the connector's supported-features table and auth notes at [Google BigQuery connector \(Power Query\)](#). Check it when you set this up for real — the details below are stable, the dialogs evolve.

Step 1: Connect from Power BI Desktop

1. **Home → Get Data → More...**, then find **Google BigQuery** (search for it, or look in the Database category) → **Connect**. The connector ships in the box; it is the same Power Query connector Excel uses.
2. Leave **Advanced options** alone for a first connection (that is where a billing project or a custom SQL statement would go). Select **OK**.
3. **Sign in.** Choose the organizational-account sign-in, then **Sign In**. A "Sign in with Google" dialog appears: pick your Google account and approve connecting to Power BI Desktop. Select **Connect**.
4. **Navigator.** Expand the tree — project → dataset → table — and pick your mart table. Projects and datasets appear exactly as they do in the GCP console: the resource hierarchy you learned on Day 1, surfacing inside your existing BI tool.
5. **Choose Import or DirectQuery** when the dialog offers both. For a first look, pick **Import** and load — the decision deserves the table below, not a coin flip.

6. Select **Load**. The table lands in the model, in the Fields pane. From here everything is the Power BI you already know — DAX, visuals, publish to the service. Nothing on the report side changes.

What the Google sign-in actually means: the connector uses OAuth, so **BigQuery sees the report author as a person**. IAM on the Gold dataset applies to each author individually — including any row- or column-level security you define there — and the BigQuery audit trail shows real people running real queries, not a shared SQL login. Access is granted and revoked in IAM, per person. That is a feature: treat it as one, and never share a single Google account across authors.

Step 2: The decision — Import vs DirectQuery

The connector supports both. They are not interchangeable; choose per mart.

	Import (data cached in Power BI)	DirectQuery (live queries to BigQuery)
Latency	Stale until the next refresh — fine for daily marts	Current on every visual interaction
BigQuery cost	One query per refresh — small and predictable	A query per visual, per user, per click — bills scan bytes continuously; on on-demand pricing this needs watching
Report performance	Fastest — data sits in-memory in Power BI	Bounded by BigQuery response time — usually seconds; fine on well-partitioned Gold tables
Features	Full DAX, all visuals	Some DAX and visual limitations (standard DirectQuery rules)
Fits Iowa when...	Default for your nightly-built Gold marts — Dataform rebuilds the marts on a schedule anyway, so a refresh right after the build matches reality	Executives need intra-day freshness, or a dataset is too large to cache

Recommendation: start with Import plus scheduled refresh, aligned to the Dataform release schedule. Your marts are rebuilt nightly by design — DirectQuery buys freshness the pipeline does not have yet. Revisit per mart when streaming use cases appear; and note the cost math changes if you move to slot/flat-rate pricing — that is a Data Warehousing course conversation.

Step 3: Publish — auth in the Power BI service

Two identities, two stories. Keep them distinct.

- **Power BI Desktop (report authors):** each author signs in with their **own Google account (OAuth)**, as above. BigQuery sees the person; IAM and row/column security apply per author; the audit trail is real people. No service accounts on desktops.
- **Power BI service (published reports, scheduled refresh):** the stored data-source credentials should be a **service account** — a non-human identity whose only job is "read the Gold dataset." Minimum roles: roles/bigquery.dataViewer on the Gold dataset, plus roles/bigquery.jobUser on the project so it can run query jobs. Set this under the semantic model's **Settings** → **Data source credentials** in the service; the connector supports service-account sign-in (account email plus JSON key). Scheduled refresh of an Import dataset runs as this identity.

You already enforce this discipline today — a service account for the Power BI service is the same instinct as a SQL Agent proxy account: a least-privilege, non-human identity that owns exactly one job. Same rule, new console.

One honest caveat: the exact credential options in the service evolve. The pattern — user OAuth on the desktop, service account in the service — is stable; confirm the current dialogs on the [Microsoft connector page](#) when you set it up.

Gateway note — for the M&O team

BigQuery is a **cloud** source: scheduled refresh from the Power BI service to BigQuery needs **no on-premises data gateway**. The gateway enters the picture only if a report **mashes up BigQuery with on-prem sources** (your SQL Server, file shares) — then one on-premises data gateway serves the on-prem legs, the same gateway infrastructure you run today. As sources move to the cloud, your gateway footprint gets quieter, not bigger.

Common gotchas

- **Signed in with the wrong Google account.** Lab accounts, personal Gmail, work identity — the OAuth dialog happily uses whichever browser session is active, and Power BI caches it. Symptom: Navigator shows the wrong projects, or none. Fix: **File** → **Options and settings** → **Data source settings**, clear the BigQuery permission, and sign in again deliberately.
- **Can't see your project or dataset in the Navigator.** That is almost always IAM, not the connector: your account needs dataViewer on the dataset and jobUser on the project. Test in the BigQuery console first — if you can query the table there, the connector will see it.
- **Wrong project picked.** Same wrong-project trap as the console picker on Day 1 — in an estate with many projects, confirm the project name before you expand anything.

- **DirectQuery bill shock.** Every visual interaction by every user issues a live BigQuery query. On on-demand pricing, a popular DirectQuery dashboard is a metering device. This is why the default above is Import.
- **Scheduled refresh fails after publish.** You loaded data in Desktop, so it must be fine — no: the service needs its own stored credentials (the service account above), and that account needs jobUser or every refresh job fails to run.
- **Google sign-in fails or loops.** Check the Power BI Desktop version — pre-June-2021 builds cannot complete Google sign-in.
- **IT policy demands a gateway.** Only true if the report also touches on-prem sources. BigQuery alone never needs one.

Verify it works

- ☐ The mart table appears in the Fields pane after Load, with the expected row count (compare against a COUNT(*) in the BigQuery console).
- ☐ A visual you already own (reuse an existing report page) renders against the new source with no measure changes.
- ☐ After publish, the semantic model's data-source credentials hold the service account, and a manual **Refresh now** succeeds in the service.
- ☐ BigQuery's job history shows your queries under your identity (desktop) and the service account's identity (service refresh) — the audit story you will be asked about.

What this replaces in your current stack

- The SSIS extract → reporting-copy tables that exist only so Power BI has something stable to read — Gold marts in BigQuery take that role.
- The ODBC DSN / SQL Server connection entries in every author's Power BI Desktop — replaced by the Google BigQuery connector and per-person OAuth.
- The shared SQL login embedded in published datasets — replaced by a least-privilege service account, with real people in the audit trail.

The closing idea

Your Gold marts land in the Power BI you already have. Same report authors, same workspaces, same DAX, same consumers — the only thing that changes is the connection string. Nothing on the report side changes. That is the whole migration, end to end: kill the extracts, land in BigQuery, Bronze/Silver/Gold with Dataform, and the reports keep working.

Part of the Iowa DOM DoIT Data Engineering student pack — see ../day2/00_schedule.md.